

Construction Safety Plan

1. Introduction

- a. This Safety Plan outlines the procedures and protocols to ensure the safe execution of [Rec Center Renovation- Phase III](#) focusing on the use of scaffolding and personal protective equipment (PPE) to mitigate risks and comply with safety regulations.
- b. The project involves safety challenges that require adherence to EM385-1 standards.

2. Project Overview

- a. The project involves repair works in leaned part of retainer wall. Work involves site preparation, demolition, debris removal, excavation, footing, concreting, masonry works, molding and deign installation and finishing.
- b. The Safety Plan addresses specific hazards associated with working at height, materials handling, and potential exposure to falling objects.

3. Responsibilities

- a. Safety Officer: [Name, Phone Number, Email](#)
 - i. Responsible for overseeing safety compliance, conducting safety audits, and coordinating safety training.
- b. Project Manager: [Name, Phone Number, Email](#)
 - i. Ensures safety procedures are integrated into project planning and execution.
- c. Scaffold Supervisor: [\[Name, Phone Number, Email\]](#)
 - i. Oversees scaffold erection, inspection, and maintenance.
- d. Workers: All personnel involved in repair woks in retainer wall
 - i. Responsible for adhering to safety procedures, wearing appropriate PPE, and reporting hazards promptly.

4. Hazard Communication

4.1 Pre-Job Hazard Analysis (PJHA):

- 4.1.1** Conducted before each shift to identify and mitigate potential hazards associated with the repair, maintenance and restoration works

4.2 Toolbox Talks:

- 4.2.1 Weekly meetings to discuss project-specific hazards, safety procedures, and emergency protocols.

4.3 Signage and Communication:

- 4.3.1 Clear signage indicating restricted areas, scaffold load capacities, and emergency contact information.

5. Site Safety and Health Plan (SSHP)

5.1 Site-Specific Hazards:

- 5.1.1 Fall hazards due to working at height, potential for dropped objects, and electrical hazards.

5.2 Emergency Procedures:

- 5.2.1 Evacuation routes, assembly points, and procedures for medical emergencies and severe weather.

5.3 Incident Reporting:

- 5.3.1 Procedures for documenting and reporting accidents, near misses, and injuries.

6. Personal Protective Equipment (PPE) – See section 15.3

6.1 PPE Assessment:

- 6.1.1 Hard hats, safety glasses, gloves, high-visibility vests, and sturdy footwear required at all times.
- 6.1.2 Fall protection equipment (harnesses, lanyards, anchor points) for workers on scaffolding.

6.2 PPE Use and Maintenance:

- 6.2.1 Workers responsible for inspecting PPE before each use, reporting defects, and proper storage after use.

7. Scaffolding Safety

7.1 Scaffold Design and Erection:

- 7.1.1 Erected and maintained by competent personnel following manufacturer guidelines and OSHA standards.

7.2 Scaffold Inspection:

- 7.2.1 Daily inspections before use and weekly inspections by a competent person to ensure structural integrity.

7.3 Load Capacity:

- 7.3.1 Scaffold load capacities posted and adhered to at all times.

8. Fall Protection Plan

8.1 Fall Hazard Assessment:

- 8.1.1 Identification of potential fall hazards, including scaffold edges, openings, and working near unprotected sides and edges of the building.

8.2 Fall Protection Systems:

- 8.2.1 Guardrails, personal fall arrest systems (PFAS), and safety nets used where applicable.
- 8.2.2 Training provided on proper use of fall protection equipment and rescue procedures.

9. Electrical Safety

9.1 Electrical Hazard Recognition:

- 9.1.1 Safety precautions when working near electrical lines or outlets, including lockout/tagout procedures.

9.2 Temporary Power:

- 9.2.1 Ground fault circuit interrupters (GFCIs) used for all electrical equipment and tools.

10. Tool and Equipment Safety

10.1 Tool Safety:

- 10.1.1 Training provided on safe operation and maintenance of power tools and equipment specific to window replacement.

10.2 Equipment Inspection:

- 10.2.1 Regular inspection and maintenance of tools to ensure safe operation and prevent accidents.

11. Fire Prevention and Protection

11.1 Fire Prevention Measures:

11.1.1 No smoking policy, proper disposal of flammable materials.

11.2 Fire Protection Equipment:

11.2.1 Fire extinguishers located strategically and inspected regularly.

12. Environmental Protection

12.1 Environmental Hazards:

12.1.1 Proper disposal of construction debris and recycling of materials where feasible.

12.2 Spill Prevention:

12.2.1 Procedures for containing and reporting spills of hazardous materials.

13. Emergency Preparedness

13.1 Emergency Response Plan:

13.1.1 Procedures for medical emergencies, fires, severe weather, and evacuation.

13.2 Emergency Drill Procedures:

13.2.1 Regular drills to practice emergency response and evacuation procedures.

14. Recordkeeping and Documentation

14.1 Incident Reporting:

14.1.1 Procedures for documenting incidents, near misses, and corrective actions.

14.2 Safety Inspections:

14.2.1 Daily scaffold inspections and weekly safety inspections conducted by the Safety Officer or designated personnel.

15. Detailed Safety Specifications

15.1 Scaffolding

15.1.1 Design and Erection:

15.1.2 Designed, built, and inspected by competent personnel trained for this task.

15.1.3 Comply with OSHA regulations, including guardrails, mid-rails, toe boards, and access ladders.

15.2 Guardrail Requirements:

15.2.1 Top rail at 42 inches (+/- 3 inches) above working surface.

15.2.2 Mid-rail halfway between top rail and platform.

15.2.3 Guardrails secured to withstand a 200-pound horizontal force.

15.2.4 Toe board at least 4 inches in height.

15.3 Personnel Safety:

15.3.1 **Head Protection** - Construction workers who are vulnerable to head injury from impact, falling or flying objects, or electrical shock and burns, must wear protective helmets while on the job.

15.3.2 **Face and Eye Protection** - Workers use appropriate eye or face protection, including side protection such as clip-on or slide-on side shields, to prevent exposure to flying debris, splash from liquid chemicals, and other hazardous materials.

15.3.3 **Hearing Protection** - When noise levels can't be reduced below specified levels of 90dB, ear protection must be provided to ensure protection for workers. Earplugs or earmuffs are often worn to protect workers from loud noises on construction sites. Devices worn in-ear must be fitted properly.

15.3.4 **Respiratory protection** - Must ensure worker should not exposure to harmful chemicals, dust, and other respiratory hazards. Make surer respiratory protections such as mask, full face mask, half face mask etc. are properly fitted to assure the adequate protection. A loose respirator may not create a proper seal, while a tight one may cause discomfort or even breathing difficulties.

15.3.5 **Hand and Arm protection** - Gloves should be available for the workers to use as PPE against cuts, abrasions, and other hand injuries. Ensure gloves are properly fitted, so that they shall provide adequate protection. Loose gloves may fall off or snag on equipment, while tight gloves may cause discomfort or even restrict movement.

15.3.6 **Legs and Foot Protection** - Protective footwear must be worn in areas where falling or rolling items, sharp or sole-piercing objects, or electrical hazards are present. Additionally, workers' feet and legs must be protected in case of explosions, static electricity, and exposure to hot substances or corrosive materials.

15.3.7 **Skin Protection** - Workers should wear gloves that are appropriate for the scope of work being performed. For example, welders or iron workers may need gloves with thermal protection to keep their hands and arms safe from materials used at extremely hot temperatures.

15.3.8 **Fall Protection** - Construction workers performing work on leading edges (or walking or working near an unprotected edge) six feet or more above ground must use fall protection, including guardrail systems, safety net systems, or by wearing personal fall arrest systems

15.3.9 **Electrical Shock Protection** - To protect workers from shock or other injuries due to these risks, proper protective equipment - EH rated safety boots, specific rubber insulating blankets, rubber insulating matting, rubber insulating covers, rubber insulating line hose, rubber insulating gloves, and rubber insulating sleeves should be provided.

15.3.10 **Visibility Clothing** - This risk should be mitigated by improving the visibility of workers to vehicle drivers and equipment operators.

15.4 **Ramps**

15.4.1 **Design and Construction:**

15.4.1.1 Ramps not exceeding a 10:1 length-to-height ratio.

15.4.1.2 Designed to support maximum weight load.

15.4.1.3 Minimum width of 42 inches with curbs to prevent falls.

15.5 **Temporary Stairs**

15.5.1 Stairs provided for structures two or more floors or more than 20 feet in height.

15.5.2 Installed as soon as possible after placement of upper floors.

15.5.3 Free of hazardous projections.

15.5.4 Hollow pans filled with wood boards or solid material before use.

15.5.5 Kept clear of debris and loose materials.

15.5.6 Temporary stairs to have a landing at least 30 inches wide for every 12 feet of vertical rise.

15.5.7 Uniform step height and width throughout any flight of stairs.

15.6 Fall Protection

15.6.1 Safety Harness and Lifelines:

- 15.6.1.1** Issued and required for work involving potential falls of 6 feet or greater.
- 15.6.1.2** Meet or exceed OSHA standards.
- 15.6.1.3** Lanyards to be a minimum of ½ inch nylon or equivalent, with a breaking strength of 5,400 pounds.
- 15.6.1.4** Lifelines secured above the point of operation, capable of supporting a minimum of 5,400 pounds.

15.7 Fire Prevention and Control

15.7.1 Precautions:

- 15.7.1.1** All fires reported regardless of severity.
- 15.7.1.2** Prevent fires caused by operations.
- 15.7.1.3** Flammable liquids not stockpiled; stored in approved containers within a flammables cabinet.
- 15.7.1.4** No smoking or open flames within 50 feet of flammable materials.
- 15.7.1.5** Accumulation of combustible trash and debris prevented.
- 15.7.1.6** Hot work program requirements applied for cutting, welding, grinding.
- 15.7.1.7** Notify EHSRM if fire suppression systems are turned off.

15.8 Job Hazard Analysis

15.8.1 Non-Routine Projects:

- 15.8.1.1** Perform a job hazard analysis for potential risk of injury.
- 15.8.1.2** Evaluate mechanics of operation and identify safety measures.
- 15.8.1.3** Collaboration with EHSRM personnel for medical surveillance and PPE requirements.

15.9 Housekeeping

15.9.1 Site Cleanliness:

- 15.9.1.1** Regular removal of trash, debris, remnant parts, and other items.
- 15.9.1.2** Avoid stockpiling and keep walk paths clear.
- 15.9.1.3** Proper storage of tools and equipment when not in use.

15.10 Electrical Safety

15.10.1 General Requirements:

- 15.10.1.1** Use UL, FM, or recognized test/listing agent approved equipment.
- 15.10.1.2** Comply with National Electrical Safety Code (NESC) or National Electrical Code (NEC).
- 15.10.1.3** UTSA Facilities and Electrical Supervisor ensure code compliance for all electrical work.
- 15.10.1.4** De-energize equipment and circuits before work starts; use protective equipment if working on energized lines.

15.10.2 Temporary Wiring:

- 15.10.2.1** Effectively grounded in accordance with NEC.
- 15.10.2.2** Enclosed and grounded switches.
- 15.10.2.3** Closable and lockable panel boards.
- 15.10.2.4** Use GFCIs for 120-volt, 15, and 20 amp outlets.

15.10.3 Overhead Lines:

- 15.10.3.1** Considered energized until confirmed otherwise and visibly grounded.
- 15.10.3.2** Coordination required for work near overhead lines.